

Detection and Mitigation of O1 Interface Attacks in O-RAN Using SIEM Pipelines

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Motivation

- DDoS attacks have reached record scale (2025–26)
- Record volumetric bursts: 31.4 Tbps (Cloudflare), 15.7 Tbps / 3.6 Bpps (Azure)
- Peak bandwidth +85% YoY, packet rate +56% — driven by IoT super-botnets of millions of device
- Cover for stealth intrusion; telecom is a top-targeted sector



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DDoS Attack Statistics 2026: Records, Hyper-Volumetric Attacks, and Aisuru

Verified 2024 to 2026 DDoS statistics from Cloudflare, NETSCOUT, Akamai, Microsoft Azure, AWS, Google Cloud, Imperva, F5 Labs, A10, Lumen, and Verisign. Every record sourced.

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Critical Apache HTTP/2 Flaw (CVE-2026-23918) Enables DoS and Potential RCE

▲ Raviv Lakshmanan May 05, 2026

Vulnerability / Server Security



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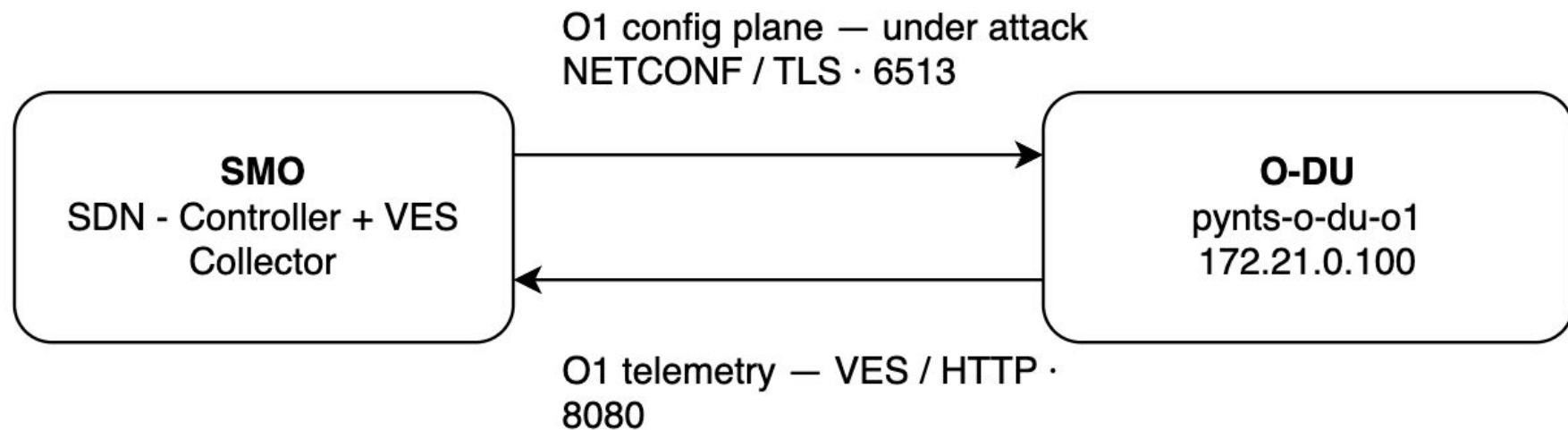
Why this matters for O-RAN

What this project targets

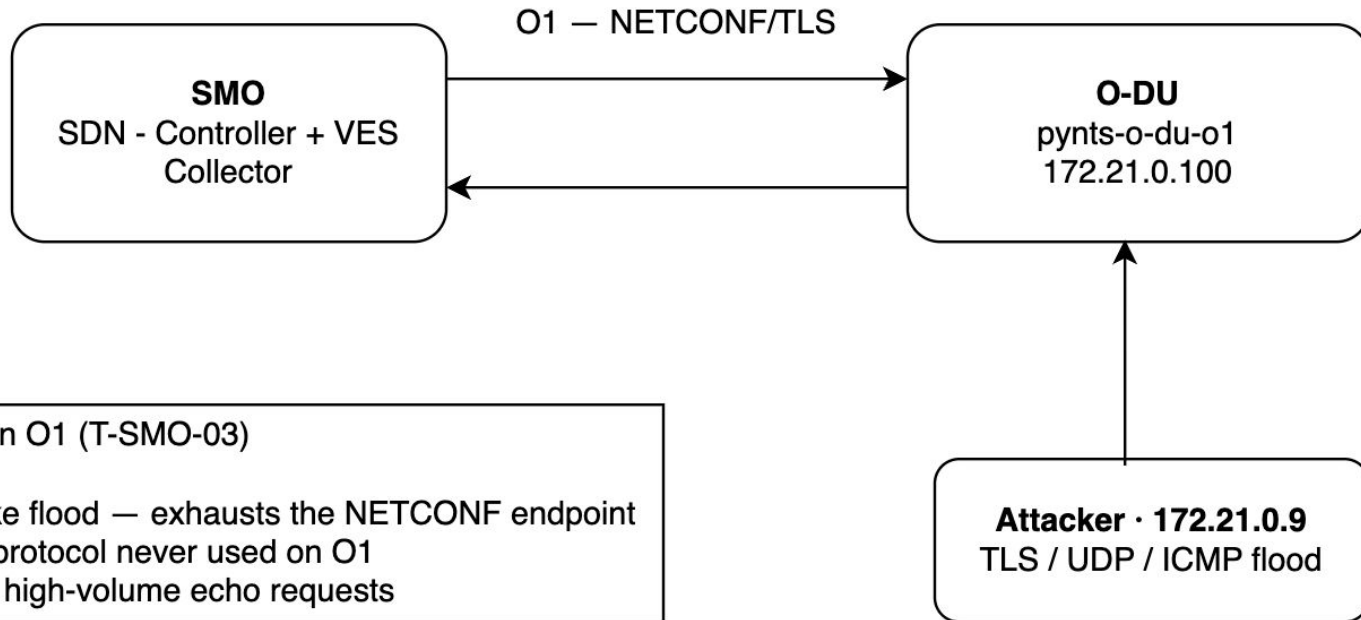
Why detection must evolve

Architecture

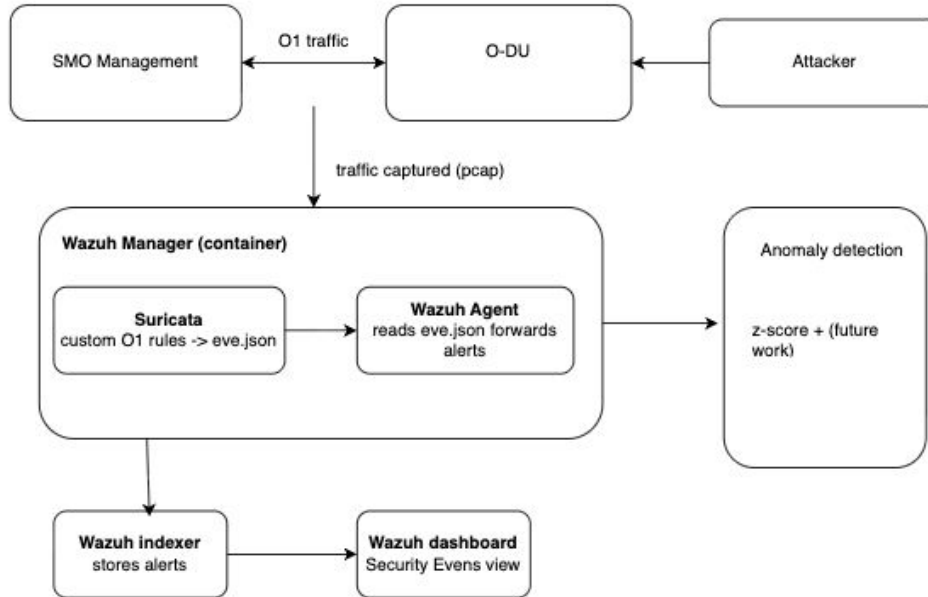
The O1 interface



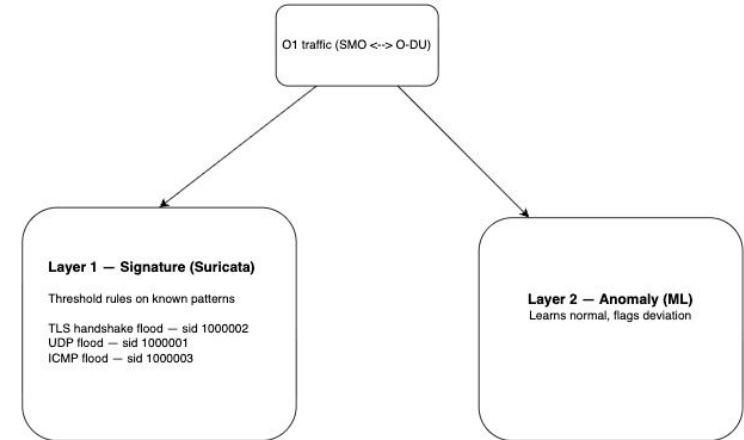
DDoS Threat Simulation



The detection pipeline



Defending the infrastructure



Monitoring stack (Wazuh 4.7.3, single-node)

- Wazuh manager, indexer, and dashboard
- Suricata installed as a process inside the Wazuh manager — the network sensor
- Runs alongside the O-RAN stack, observing its traffic without being part of it



Methodology

ICMP Flood

```
(base) eahtex86@wlan-l3025 Seminar % docker run --rm \
--net=container:pynts-o-du-01 \
-v "$PWD/captures:/caps" \
nicolaka/netshoot \
tcpdump -i any -w /caps/icmp_flood.pcap
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
^C1888856 packets captured
29858281 packets received by filter
27969425 packets dropped by kernel
(base) eahtex86@wlan-l3025 Seminar %
```

```
(base) eahtex86@wlan-l3025 Seminar % docker run --rm --net=dcn 01-attacker \
hping3 -I --flood -d 1200 172.21.0.7
hping in flood mode, no replies will be shown
^C
172.21.0.7 hping statistic --
14929128 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
HPING 172.21.0.7 (eth0 172.21.0.7): icmp mode set, 28 headers + 1200 data bytes
(base) eahtex86@wlan-l3025 Seminar %
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	be:61:b7:8d:39:7f		48	ARP	Announcement for 172.21.0.4
2	0.000148	2001:db8:1:50::4	ff02::1	ICMPv6	92	Neighbor Advertisement 2001:db8:1:50::4 (ovr) is at be:61:b7:8d:39:7f
3	0.001723	::	ff02::16	ICMPv6	116	Multicast Listener Report Message v2
4	0.063875	be:61:b7:8d:39:7f		48	ARP	Who has 172.21.0.7? Tell 172.21.0.4
5	0.063883	92:0f:7b:5a:c7:07		48	ARP	172.21.0.7 is at 92:0f:7b:5a:c7:07
6	0.063930	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=0/0, ttl=64 (reply in 7)
7	0.063968	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=0/0, ttl=64 (request in 6)
8	0.063994	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=256/1, ttl=64 (reply in 9)
9	0.063996	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=256/1, ttl=64 (request in 8)
10	0.064000	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=512/2, ttl=64 (reply in 11)
11	0.064000	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=512/2, ttl=64 (request in 10)
12	0.064004	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=768/3, ttl=64 (reply in 13)
13	0.064005	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=768/3, ttl=64 (request in 12)
14	0.064008	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=1024/4, ttl=64 (reply in 15)
15	0.064008	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=1024/4, ttl=64 (request in 14)
16	0.064011	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=1280/5, ttl=64 (reply in 17)
17	0.064016	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=1280/5, ttl=64 (request in 16)
18	0.064039	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=1536/6, ttl=64 (reply in 19)
19	0.064040	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=1536/6, ttl=64 (request in 18)
20	0.064043	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=1792/7, ttl=64 (reply in 21)
21	0.064044	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=1792/7, ttl=64 (request in 20)
22	0.064047	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=2048/8, ttl=64 (reply in 23)
23	0.064047	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=2048/8, ttl=64 (request in 22)
24	0.064050	172.21.0.4	172.21.0.7	ICMP	1248	Echo (ping) request id=0x0100, seq=2304/9, ttl=64 (reply in 25)
25	0.064051	172.21.0.7	172.21.0.4	ICMP	1248	Echo (ping) reply id=0x0100, seq=2304/9, ttl=64 (request in 24)

```
> Frame 1: Packet, 48 bytes on wire (384 bits), 48 bytes captured (384 bits)
> Linux cooked capture v2
> Address Resolution Protocol (ARP Announcement)
```

```
0000 08 06 00 00 00 00 00 00 00 01 01 06 be 61 b7 8d .....a..
0010 3f 7f 00 00 01 08 00 06 04 00 01 be 61 b7 8d .....a..
0020 3f 7f ac 15 00 04 00 00 00 00 00 ac 15 00 04 9.....
```

UDP Flood

```
(base) eahtex86@wlan-l3025 Seminar % docker run --rm \
--net=container:pynts-o-du-o1 \
-v "$PWD/captures:/caps" \
nicolaka/netshoot \
tcpdump -i any -w /caps/udp_flood.pcap
tcpdump: WARNING: any: That device doesn't support promiscuous mode
(Promiscuous mode not supported on the "any" device)
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144
bytes
^C2820572 packets captured
25378040 packets received by filter
22557468 packets dropped by kernel
(base) eahtex86@wlan-l3025 Seminar %
```

```
(base) eahtex86@wlan-l3025 Seminar % docker run --rm --net=dcn o1-attacker \
hping3 --udp --flood -d 1200 -p 6513 172.21.0.7
hping in flood mode, no replies will be shown
^C
-- 172.21.0.7 hping statistic --
25377926 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
HPING 172.21.0.7 (eth0 172.21.0.7): udp mode set, 28 headers + 1200 data byt
es
(base) eahtex86@wlan-l3025 Seminar %
```

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	3e:6d:16:2f:a1:67		ARP	48	ARP Announcement for 172.21.0.4
2	0.000150	2001:db8:1:50::4	ff02::1	ICMPv6	92	Neighbor Advertisement 2001:db8:1:50::4 (ovr) is at 3e:6d:16:2f:a1:67
3	0.001686	::	ff02::16	ICMPv6	116	Multicast Listener Report Message v2
4	0.044056	3e:6d:16:2f:a1:67		ARP	48	Who has 172.21.0.7? Tell 172.21.0.4
5	0.044063	92:0f:7b:5a:c7:07		ARP	48	172.21.0.7 is at 92:0f:7b:5a:c7:07
6	0.044104	172.21.0.4	172.21.0.7	UDP	1248	1908 - 6513 Len=1200
7	0.044557	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
8	0.044606	172.21.0.4	172.21.0.7	UDP	1248	1909 - 6513 Len=1200
9	0.044615	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
10	0.044628	172.21.0.4	172.21.0.7	UDP	1248	1910 - 6513 Len=1200
11	0.044637	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
12	0.044655	172.21.0.4	172.21.0.7	UDP	1248	1911 - 6513 Len=1200
13	0.044662	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
14	0.044679	172.21.0.4	172.21.0.7	UDP	1248	1912 - 6513 Len=1200
15	0.044687	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
16	0.044707	172.21.0.4	172.21.0.7	UDP	1248	1913 - 6513 Len=1200
17	0.044716	172.21.0.7	172.21.0.4	ICMP	596	Destination unreachable (Port unreachable)
18	0.044737	172.21.0.4	172.21.0.7	UDP	1248	1914 - 6513 Len=1200
19	0.044755	172.21.0.4	172.21.0.7	UDP	1248	1915 - 6513 Len=1200
20	0.044776	172.21.0.4	172.21.0.7	UDP	1248	1916 - 6513 Len=1200
21	0.044797	172.21.0.4	172.21.0.7	UDP	1248	1917 - 6513 Len=1200
22	0.044818	172.21.0.4	172.21.0.7	UDP	1248	1918 - 6513 Len=1200
23	0.044838	172.21.0.4	172.21.0.7	UDP	1248	1919 - 6513 Len=1200
24	0.044856	172.21.0.4	172.21.0.7	UDP	1248	1920 - 6513 Len=1200
25	0.044873	172.21.0.4	172.21.0.7	UDP	1248	1921 - 6513 Len=1200

- > Frame 1: Packet, 48 bytes on wire (384 bits), 48 bytes captured (384 bits)
- > Linux cooked capture v2
- > Address Resolution Protocol (ARP Announcement)

```
0000 08 06 00 00 00 00 00 0b 00 01 01 06 3e 6d 16 2f .....>>>/
0010 a1 67 00 00 00 01 08 00 06 04 00 01 3e 6d 16 2f .....>>>/
0020 a1 67 ac 15 00 04 00 00 00 00 00 ac 15 00 04 .....g.....
```

TL S Flood

SYNs – ~6,500

No.	Time	Source	Destination	Protocol	Length	Info
118	2471.631916	172.21.0.9	172.21.0.7	TCP	80	43922 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849425 Tscrn=65126
185	2471.631941	172.21.0.9	172.21.0.7	TCP	80	43934 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849425 Tscrn=65126
118	2471.632023	172.21.0.9	172.21.0.7	TCP	80	43944 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849425 Tscrn=65126
117	2471.632038	172.21.0.9	172.21.0.7	TCP	80	43954 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849425 Tscrn=65126
116	2471.632078	172.21.0.9	172.21.0.7	TCP	80	43959 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849425 Tscrn=65126
119	2471.632078	172.21.0.9	172.21.0.7	TCP	80	43970 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
122	2471.633193	172.21.0.9	172.21.0.7	TCP	80	43978 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
119	2471.633508	172.21.0.9	172.21.0.7	TCP	80	44084 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
124	2471.633702	172.21.0.9	172.21.0.7	TCP	80	44022 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
125	2471.633926	172.21.0.9	172.21.0.7	TCP	80	44028 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
126	2471.633971	172.21.0.9	172.21.0.7	TCP	80	44036 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
127	2471.633982	172.21.0.9	172.21.0.7	TCP	80	43992 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849426 Tscrn=65126
128	2471.633919	172.21.0.9	172.21.0.7	TCP	80	44040 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
119	2471.633970	172.21.0.9	172.21.0.7	TCP	80	44044 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
136	2471.634019	172.21.0.9	172.21.0.7	TCP	80	44048 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
140	2471.634095	172.21.0.9	172.21.0.7	TCP	80	44064 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
147	2471.634434	172.21.0.9	172.21.0.7	TCP	80	44060 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
145	2471.634524	172.21.0.9	172.21.0.7	TCP	80	44080 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849427 Tscrn=65126
146	2471.634508	172.21.0.9	172.21.0.7	TCP	80	44096 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
147	2471.634672	172.21.0.9	172.21.0.7	TCP	80	44108 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
148	2471.634835	172.21.0.9	172.21.0.7	TCP	80	44122 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
149	2471.634874	172.21.0.9	172.21.0.7	TCP	80	44126 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
158	2471.634944	172.21.0.9	172.21.0.7	TCP	80	44140 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
151	2471.634935	172.21.0.9	172.21.0.7	TCP	80	44140 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
154	2471.634934	172.21.0.9	172.21.0.7	TCP	80	44158 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
153	2471.634922	172.21.0.9	172.21.0.7	TCP	80	44178 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
154	2471.634948	172.21.0.9	172.21.0.7	TCP	80	44162 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
155	2471.634989	172.21.0.9	172.21.0.7	TCP	80	44190 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
157	2471.634918	172.21.0.9	172.21.0.7	TCP	80	44204 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
157	2471.634943	172.21.0.9	172.21.0.7	TCP	80	44222 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
158	2471.633986	172.21.0.9	172.21.0.7	TCP	80	44224 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
159	2471.633991	172.21.0.9	172.21.0.7	TCP	80	44232 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
160	2471.633922	172.21.0.9	172.21.0.7	TCP	80	44240 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
161	2471.633926	172.21.0.9	172.21.0.7	TCP	80	44242 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
162	2471.633932	172.21.0.9	172.21.0.7	TCP	80	44254 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
163	2471.633920	172.21.0.9	172.21.0.7	TCP	80	44266 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
164	2471.633925	172.21.0.9	172.21.0.7	TCP	80	44274 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849428 Tscrn=65126
165	2471.633918	172.21.0.9	172.21.0.7	TCP	80	44286 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
166	2471.633946	172.21.0.9	172.21.0.7	TCP	80	44300 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
167	2471.633974	172.21.0.9	172.21.0.7	TCP	80	44308 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
168	2471.633976	172.21.0.9	172.21.0.7	TCP	80	44312 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
169	2471.633970	172.21.0.9	172.21.0.7	TCP	80	44320 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
170	2471.633940	172.21.0.9	172.21.0.7	TCP	80	44328 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126
171	2471.633913	172.21.0.9	172.21.0.7	TCP	80	44330 → 6513 [SYN] Seq=6164240 Len=0 MSS=1460 SACK_PERM TSval=176849429 Tscrn=65126

```

0000 00 00 00 00 00 00 00 00 00 01 04 05 52 07 7b 51 .....
Linux cooked capture v2
Internet Protocol Version 4, Src: 172.21.0.9, Dst: 172.21.0.7
Transmission Control Protocol, Src Port: 43970, Dst Port: 6513, Seq #: 1, Len #: 8

```

Packets: 10733 (Displayed: 640 (6.0%))

SYN-ACKs – ~950

No.	Time	Source	Destination	Protocol	Length	Info
186	2471.651945	172.21.0.7	172.21.0.9	TCP	80	6513 → 45922 [SYN, ACK] Seq=61647584 Len=0 MSS=1460 SACK_PERM TSval=1146674583 Tscrn=176849425 WS=128
117	2471.651955	172.21.0.7	172.21.0.9	TCP	80	6513 → 45934 [SYN, ACK] Seq=61647583 Len=0 MSS=1460 SACK_PERM TSval=1146674583 Tscrn=176849425 WS=128
107	2471.652035	172.21.0.7	172.21.0.9	TCP	80	6513 → 45944 [SYN, ACK] Seq=61647584 Len=0 MSS=1460 SACK_PERM TSval=1146674583 Tscrn=176849425 WS=128
114	2471.652061	172.21.0.7	172.21.0.9	TCP	80	6513 → 45954 [SYN, ACK] Seq=61647583 Len=0 MSS=1460 SACK_PERM TSval=1146674583 Tscrn=176849425 WS=128
117	2471.652299	172.21.0.7	172.21.0.9	TCP	80	6513 → 45958 [SYN, ACK] Seq=61647583 Len=0 MSS=1460 SACK_PERM TSval=1146674583 Tscrn=176849425 WS=128
120	2471.652844	172.21.0.7	172.21.0.9	TCP	80	6513 → 45970 [SYN, ACK] Seq=61647584 Len=0 MSS=1460 SACK_PERM TSval=1146674584 Tscrn=176849426 WS=128
193	2471.667781	172.21.0.7	172.21.0.9	TCP	80	6513 → 46470 [SYN, ACK] Seq=61647519 Len=0 MSS=1460 SACK_PERM TSval=1146674519 Tscrn=176849441 WS=128
194	2471.667787	172.21.0.7	172.21.0.9	TCP	80	6513 → 46498 [SYN, ACK] Seq=61647519 Len=0 MSS=1460 SACK_PERM TSval=1146674519 Tscrn=176849441 WS=128
195	2471.667780	172.21.0.7	172.21.0.9	TCP	80	6513 → 46482 [SYN, ACK] Seq=61647519 Len=0 MSS=1460 SACK_PERM TSval=1146674519 Tscrn=176849441 WS=128
196	2471.667788	172.21.0.7	172.21.0.9	TCP	80	6513 → 46484 [SYN, ACK] Seq=61647519 Len=0 MSS=1460 SACK_PERM TSval=1146674519 Tscrn=176849441 WS=128
275	2471.952802	172.21.0.7	172.21.0.9	TCP	80	6513 → 46854 [SYN, ACK] Seq=616474776 Len=0 MSS=1460 SACK_PERM TSval=1146674776 Tscrn=176849698 WS=128
281	2471.952283	172.21.0.7	172.21.0.9	TCP	80	6513 → 46856 [SYN, ACK] Seq=616474883 Len=0 MSS=1460 SACK_PERM TSval=1146674883 Tscrn=176849725 WS=128
299	2472.084109	172.21.0.7	172.21.0.9	TCP	80	6513 → 46858 [SYN, ACK] Seq=616474935 Len=0 MSS=1460 SACK_PERM TSval=1146674935 Tscrn=176849857 WS=128
311	2472.134615	172.21.0.7	172.21.0.9	TCP	80	6513 → 46874 [SYN, ACK] Seq=616474988 Len=0 MSS=1460 SACK_PERM TSval=1146674988 Tscrn=176849920 WS=128
323	2472.167990	172.21.0.7	172.21.0.9	TCP	80	6513 → 46876 [SYN, ACK] Seq=616475018 Len=0 MSS=1460 SACK_PERM TSval=1146675018 Tscrn=176849941 WS=128
338	2472.302912	172.21.0.7	172.21.0.9	TCP	80	6513 → 46892 [SYN, ACK] Seq=616475194 Len=0 MSS=1460 SACK_PERM TSval=1146675194 Tscrn=176849876 WS=128
350	2472.370666	172.21.0.7	172.21.0.9	TCP	80	6513 → 46908 [SYN, ACK] Seq=616475222 Len=0 MSS=1460 SACK_PERM TSval=1146675222 Tscrn=176849144 WS=128
361	2472.387872	172.21.0.7	172.21.0.9	TCP	80	6513 → 46918 [SYN, ACK] Seq=616475239 Len=0 MSS=1460 SACK_PERM TSval=1146675239 Tscrn=176849181 WS=128
371	2472.523207	172.21.0.7	172.21.0.9	TCP	80	6513 → 46934 [SYN, ACK] Seq=616475374 Len=0 MSS=1460 SACK_PERM TSval=1146675374 Tscrn=176849720 WS=128
383	2472.595331	172.21.0.7	172.21.0.9	TCP	80	6513 → 46918 [SYN, ACK] Seq=616475308 Len=0 MSS=1460 SACK_PERM TSval=1146675308 Tscrn=176849362 WS=128
395	2472.602327	172.21.0.7	172.21.0.9	TCP	80	6513 → 46926 [SYN, ACK] Seq=616475453 Len=0 MSS=1460 SACK_PERM TSval=1146675453 Tscrn=176849375 WS=128
444	2472.760688	172.21.0.7	172.21.0.9	TCP	80	[TCP Retransmission] 6513 → 46484 [SYN, ACK] Seq=616475508 Len=0 MSS=1460 SACK_PERM TSval=1146675508 Tscrn=176849758 WS=128
580	2472.739872	172.21.0.7	172.21.0.9	TCP	80	6513 → 46930 [SYN, ACK] Seq=616475590 Len=0 MSS=1460 SACK_PERM TSval=1146675590 Tscrn=176849512 WS=128
513	2472.869281	172.21.0.7	172.21.0.9	TCP	80	6513 → 46932 [SYN, ACK] Seq=616475608 Len=0 MSS=1460 SACK_PERM TSval=1146675608 Tscrn=176849570 WS=128
525	2472.818095	172.21.0.7	172.21.0.9	TCP	80	6513 → 46944 [SYN, ACK] Seq=616475658 Len=0 MSS=1460 SACK_PERM TSval=1146675658 Tscrn=176849595 WS=128
538	2472.954047	172.21.0.7	172.21.0.9	TCP	80	6513 → 46948 [SYN, ACK] Seq=616475686 Len=0 MSS=1460 SACK_PERM TSval=1146675686 Tscrn=176849720 WS=128
550	2473.021383	172.21.0.7	172.21.0.9	TCP	80	6513 → 46954 [SYN, ACK] Seq=616475708 Len=0 MSS=1460 SACK_PERM TSval=1146675708 Tscrn=176849795 WS=128
562	2473.023945	172.21.0.7	172.21.0.9	TCP	80	6513 → 46964 [SYN, ACK] Seq=616475781 Len=0 MSS=1460 SACK_PERM TSval=1146675781 Tscrn=176849802 WS=128
576	2473.177413	172.21.0.7	172.21.0.9	TCP	80	6513 → 46972 [SYN, ACK] Seq=616475829 Len=0 MSS=1460 SACK_PERM TSval=1146675829 Tscrn=176849951 WS=128
586	2473.235652	172.21.0.7	172.21.0.9	TCP	80	6513 → 46978 [SYN, ACK] Seq=616475847 Len=0 MSS=1460 SACK_PERM TSval=1146675847 Tscrn=176849900 WS=128
599	2473.246585	172.21.0.7	172.21.0.9	TCP	80	6513 → 46992 [SYN, ACK] Seq=616475908 Len=0 MSS=1460 SACK_PERM TSval=1146675908 Tscrn=176849120 WS=128
609	2473.392676	172.21.0.7	172.21.0.9	TCP	80	6513 → 46996 [SYN, ACK] Seq=616476024 Len=0 MSS=1460 SACK_PERM TSval=1146676024 Tscrn=176849166 WS=128
621	2473.435867	172.21.0.7	172.21.0.9	TCP	80	6513 → 47012 [SYN, ACK] Seq=616476383 Len=0 MSS=1460 SACK_PERM TSval=1146676383 Tscrn=1768491225 WS=128
633	2473.463801	172.21.0.7	172.21.0.9	TCP	80	6513 → 47028 [SYN, ACK] Seq=616476333 Len=0 MSS=1460 SACK_PERM TSval=1146676333 Tscrn=1768491235 WS=128
646	2473.433652	172.21.0.7	172.21.0.9	TCP	80	6513 → 47038 [SYN, ACK] Seq=616476444 Len=0 MSS=1460 SACK_PERM TSval=1146676444 Tscrn=1768491386 WS=128
658	2473.674665	172.21.0.7	172.21.0.9	TCP	80	6513 → 47046 [SYN, ACK] Seq=616476508 Len=0 MSS=1460 SACK_PERM TSval=1146676508 Tscrn=1768491447 WS=128
668	2473.681882	172.21.0.7	172.21.0.9	TCP	80	6513 → 47058 [SYN, ACK] Seq=616476558 Len=0 MSS=1460 SACK_PERM TSval=1146676558 Tscrn=1768491454 WS=128
774	2473.834821	172.21.0.7	172.21.0.9	TCP	80	6513 → 47062 [SYN, ACK] Seq=616476683 Len=0 MSS=1460 SACK_PERM TSval=1146676683 Tscrn=1768491807 WS=128
785	2473.894667	172.21.0.7	172.21.0.9	TCP	80	6513 → 47070 [SYN, ACK] Seq=616476745 Len=0 MSS=1460 SACK_PERM TSval=1146676745 Tscrn=1768491687 WS=128
795	2473.903360					

The SIEM detection pipeline with signature-based detection

Pipeline overview

- Goal: turn raw O1 attack traffic into actionable alerts an analyst can see
- Four stages: capture → detect (Suricata) → forward (Wazuh agent) → store & visualise (indexer + dashboard)
- Suricata runs as a process *inside* the Wazuh manager

Traffic ingestion

- Attack and O1 traffic captured to pcap on the O-DU's dcn interface
- `suricata -r <pcap>`

Suricata — detection engine

- Reads the pcap and matches traffic against a ruleset
- Custom rules, stored in `oran-custom.rules`, registered under `suricata.yaml`
- Alerts written as JSON to `eve.json`

Three custom signatures:

- UDP flood

alert udp any any -> any 6513 (msg:"O1 Interface UDP Flood - T-SMO"; threshold: type both, track by_dst, count 100, seconds 1; sid:1000001; rev:1;)

- TLS handshake flood

alert tcp any any -> any 6513 (msg:"ORAN_T-SMO TLS handshake flood detected"; flow:to_server; content:"|16 03|"; depth:2; threshold: type threshold, track by_src, count 100, seconds 5; classtype:attempted-dos; sid:1000002; rev:1;)

- ICMP flood

alert icmp 172.21.0.0/16 any -> 172.21.0.7 any (msg:"ORAN T-SMO-03 ICMP flood on O1 interface O-DU"; itype:8; dsize:>1000; threshold: type threshold, track by_src, count 100, seconds 1; classtype:attempted-dos;sid:1000003;rev:1;)

Wazuh Dashboard Results

TLS Detection

Security Alerts

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Aug 12, 2026 @ 14:14:07.434	000	wazuh.manager			Suricata: Alert - ORAN_T-SMO TLS handshake flood detected	3	86601
> Aug 12, 2026 @ 14:14:07.385	000	wazuh.manager			Suricata: Alert - ORAN_T-SMO TLS handshake flood detected	3	86601
> Aug 12, 2026 @ 14:14:07.377	000	wazuh.manager			Suricata: Alert - ORAN_T-SMO TLS handshake flood detected	3	86601

ICMP Detection

Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Aug 28, 2026 @ 12:32:30.063	000	wazuh.manager			Suricata: Alert - ORAN T-SMO-03 ICMP flood on O1 interface O-DU	3	86601
> Aug 28, 2026 @ 12:32:30.063	000	wazuh.manager			Suricata: Alert - ORAN T-SMO-03 ICMP flood on O1 interface O-DU	3	86601
> Aug 28, 2026 @ 12:32:30.062	000	wazuh.manager			Suricata: Alert - ORAN T-SMO-03 ICMP flood on O1 interface O-DU	3	86601

UDP Detection

Security Alerts

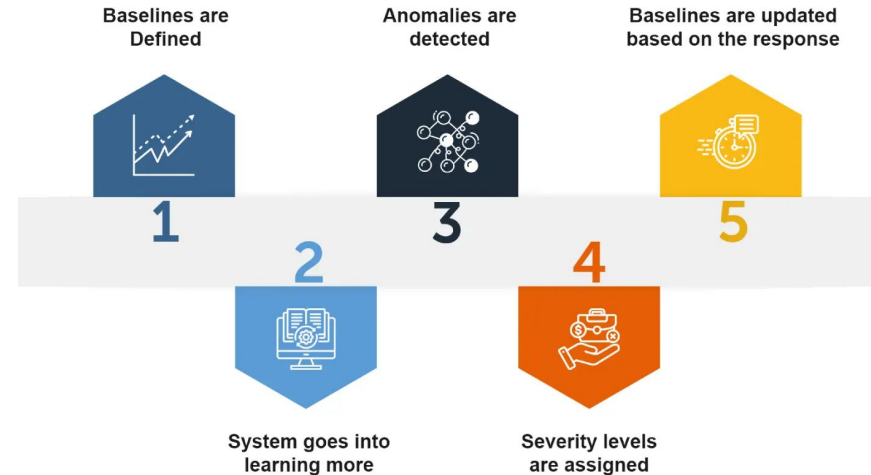
Time ↓	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
> Jul 30, 2026 @ 10:03:27.300	000	wazuh.manager			Suricata: Alert - O1 Interface UDP Flood - T-SMO-03	3	86601
> Jul 30, 2026 @ 10:03:27.300	000	wazuh.manager			Suricata: Alert - O1 Interface UDP Flood - T-SMO-03	3	86601
> Jul 30, 2026 @ 10:03:27.300	000	wazuh.manager			Suricata: Alert - O1 Interface UDP Flood - T-SMO-03	3	86601

Anomaly-based detection with z-score

Why anomaly detection, and the need for a baseline

- Signature detection catches known patterns; anomaly detection catches deviation from normal
- Unsupervised approach: learn what normal O1 traffic looks like, then flag anything statistically far from it
- Requires a **benign baseline** — a dataset of normal traffic, separate from any attack

How Does Anomaly-Based IDS Work?



Capturing the benign baseline

- Traffic made **irregular on purpose**
- Captured on the O-DU's dcn interface (eth0)
- ~20 minutes of active traffic
- Confirmed content: TLS-dominated, sparse ICMP, no UDP

Feature extraction — time-windowed features

- Unit of analysis: 1-second windows
- Features include: packets/sec, bytes/sec, mean & standard-deviation of packet size, unique destination ports, **new-connection count (SYNs)**, **TLS handshake count**, per-protocol fractions
- The discriminating features for TLS: `syn_count` and `tls_handshake` — near-zero in benign (one persistent session), high in the flood (a new connection per handshake)

The z-score model and evaluation

- **Method**
- Formula: $z = (x - \mu) / \sigma$;
- **Train/test split:** first 80% for training, last 20% for testing
- **Evaluated on three sets:** held-out benign (false positives), TLS attack (detection), and — as a cross-check — ICMP attack
- **Results:** TLS flood detected at ~71%, ICMP at 100%, false-positive rate ~6.6% on benign

Future Work

- Machine-learning anomaly detection
- Richer benign baseline
- Broader threat coverage
- Live deployment and validation
- Automated mitigation (closing the loop)

Questions ?